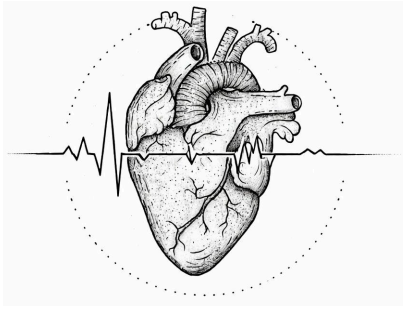


COMP721: MACHINE LEARNING (2026)

Cardiovascular Disease Risk Prediction

Using Classical Machine Learning, Ensemble Learning,

and Explainable Artificial Intelligence

Module Code	COMP721
Module Name	Machine Learning
Project Title	Cardiovascular Disease Risk Prediction Using Classical Machine Learning, Ensemble Learning, and Explainable Artificial Intelligence
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Abstract

Cardiovascular disease (CVD) remains the primary cause of mortality globally, emphasising the urgent need for reliable early risk prediction systems. While machine learning (ML) offers promising tools for risk stratification, clinical adoption is often limited by model opacity, insufficient feature engineering, and a lack of rigorous statistical validation when comparing complex ensembles to individual classifiers [9].

This project presents an end-to-end, interpretable machine learning pipeline for cardiovascular disease prediction utilising the Kaggle Cardiovascular Disease dataset. The methodology encompasses comprehensive data preprocessing, data cleaning, and the creation of eight engineered features, including clinically meaningful indicators (Body Mass Index, hypertension indicator, and Pulse Pressure), to enrich feature representation. Following feature selection, two foundational classifiers, a Decision Tree and a Support Vector Machine (SVM), were trained alongside a Soft Voting Ensemble that aggregates their predicted class probability distributions. Each baseline model underwent hyperparameter optimisation across four distinct configurations using 5-fold cross-validation on the training set.

To bridge the interpretability gap, SHapley Additive Explanations (SHAP) were integrated to elucidate global feature importance and deliver local, patient-level explanations. Furthermore, rigorous statistical hypothesis testing using paired t-tests (with Wilcoxon signed-rank tests as the fallback where normality was rejected) at $\alpha = 0.05$ was conducted across cross-validation folds to establish whether the Soft Voting Ensemble achieved statistically significant performance gains over the individual Decision Tree and SVM models [9].

The selected 10-feature set maintained predictive performance, and the Soft Voting Ensemble achieved the highest test accuracy (73.89%) and ROC-AUC (0.8015). It was not significantly different from the Decision Tree, but significantly outperformed the SVM in cross-validated accuracy ($p = 0.0099$) and F1-score ($p = 0.0078$). SHAP analysis identified systolic blood pressure, hypertension, age, and cholesterol as the main drivers of predictions, supporting a transparent framework for early cardiovascular intervention.

1. Introduction

Cardiovascular Disease (CVD) describes a wide range of conditions affecting the heart and blood vessels, commonly involving the narrowing or blockage of arteries. These heart disorders, including coronary artery disease, heart failure, and arrhythmias, often malfunction the heart's pumping systems. This leads to life threatening medical events such as heart attack and strokes [5].

CVD is the leading cause of mortality worldwide. It is estimated that CVD is responsible for 17.9 to 18 million deaths per annum, representing approximately 32% of all global fatalities [6][8]. Specifically in South Africa, CVD claims more lives than all cancers combined, second to HIV/AIDS. To emphasise the seriousness, every hour in South Africa 5 people have heart attacks, and 10 people have strokes. This escalating crisis places a massive socioeconomic burden on healthcare systems, families, and governments [10].

The prevalence of CVD is driven by a complex interaction of physiological and behavioural risk factors. Major contributors include: [1][10]

1. Hypertension (High Blood Pressure) which is the leading global risk factor for CVD death. South Africa has the highest reported rate for people aged 50 and over

2. Obesity and Unhealthy Diet. South Africa has one of the highest levels of overweight and obesity globally, often linked to high salt and fat intake.

3. Lifestyle Choices such as Tobacco use, physical inactivity, harmful alcohol consumption, and chronic stress.

Early detection is extremely important because up to 80% of premature deaths (deaths before age 65) can be prevented through timely lifestyle interventions and medical management. Identifying high-risk individuals through automated predictive systems allows healthcare professionals to begin counselling and treatment early, which significantly reduces the risk of heart failure and increases recovery rates.

An existing solution is traditional diagnosis which relies on clinical examinations, ECGs, laboratory tests, and angiography, which are expensive, invasive and unavailable in resource limited regions [2][8][9]. Recently, machine learning (ML) has emerged as a powerful tool, with researchers exploring supervised learning algorithms such as Logistic Regression, K-Nearest Neighbours (KNN), Naive Bayes, Support Vector Machines (SVM), and Random Forests [3][4][7]. With these tools comes the black box problem, where models lack interpretability, which make it difficult for clinicians to understand or trust the outputs, and data issues such as an overreliance on single models, small datasets with little diversity [2][9].

While many studies report high accuracy, they often fail to address the real-world applicability of these models in clinical settings. There is limited research that successfully integrates high predictive performance with explainability frameworks. Additionally, many studies lack a detailed statistical comparison between individual classifiers and ensemble models to determine if combining algorithms actually provides a significant benefit [8].

This project will develop a complete end-to-end machine learning pipeline using the Kaggle Cardiovascular Disease dataset. The methodology includes Feature Engineering (Creating eight new features) to improve predictive depth, Ensemble Learning (Developing and comparing a Decision Tree and an SVM against an Ensemble Model), Explainable AI (XAI) (Implementing SHAP to provide global and local explanations for model predictions, ensuring they align with medical knowledge) and Statistical Validation (Using paired t-tests or Wilcoxon signed-rank tests to confirm if the ensemble approach statistically outperforms individual models).

This study is valuable because it moves beyond black box predictions to create interpretable and reliable AI-driven diagnostics. By balancing high accuracy with SHAP explanations, the research aims to foster clinician trust and ease the adoption of AI into clinical workflows. Ultimately, this contributes to more efficient risk stratification, supporting medical practitioners in saving lives through evidence-based, early intervention [9].

2. Literature Review

Cardiovascular disease (CVD) remains one of the leading causes of morbidity and mortality worldwide and represents a significant public health challenge. CVD accounts for approximately 32% of global deaths, with more than 75% of cardiovascular deaths occurring in low- and middle-income countries [11], [12]. Its development is influenced by a combination of non-modifiable factors, including age, sex, ethnicity, and family history, and modifiable factors such as smoking, hypertension, elevated cholesterol, diabetes, obesity, physical inactivity, and stress [12], [13]. Since CVD can develop silently before symptoms become apparent, early risk prediction is important for enabling preventive interventions and improving patient outcomes [11]. Accurate risk prediction can also assist healthcare systems in prioritising high-risk individuals and allocating limited clinical resources more effectively [11], [13].

Traditional cardiovascular risk prediction methods, including models based on logistic regression and Cox proportional hazards models, have historically been used to estimate cardiovascular risk [15]. Although these approaches provide interpretable and computationally efficient benchmarks, they can have difficulty representing the complex, non-linear relationships and interactions present in multidimensional electronic health record (EHR) data [11], [15]. Consequently, machine learning (ML) and deep learning (DL) approaches have increasingly been investigated as

alternatives for CVD risk prediction. Recent evidence suggests that ensemble and deep learning approaches can achieve strong predictive performance. In an EHR-based meta-analysis, Random Forest achieved the highest pooled AUC of 0.865 (95% CI: 0.812 - 0.917), while deep learning achieved a pooled AUC of 0.847 (95% CI: 0.766 - 0.927), compared with 0.796 for logistic regression [15]. These findings demonstrate the potential of ML to capture complex patterns that may not be adequately represented by conventional statistical approaches.

Several ML algorithms have been applied to CVD prediction. Logistic Regression remains an important baseline because of its simplicity, transparency, and low computational requirements [15]. Random Forest is particularly effective for high-dimensional and non-linear data and provides useful measures of feature importance [12]. Support Vector Machines have also demonstrated competitive performance, although their application to large-scale longitudinal EHR data can be more challenging [12], [14]. XGBoost has become another widely investigated approach because of its strong predictive performance and scalability. For example, one comparative study reported an XGBoost model achieving an accuracy of 97.57%, sensitivity of 96.61%, and AUC of 0.98 after addressing class imbalance using SMOTE [13]. However, the substantially lower pooled XGBoost AUC of 0.776 reported in the broader EHR meta-analysis illustrates that model performance can vary considerably depending on the dataset, preprocessing, and evaluation methodology [15]. DL approaches have also shown considerable potential, with studies reporting DL models outperforming conventional ML models in seven of eight direct comparisons for survival outcomes [12]. Models such as DeepSurv and DeepHit are particularly relevant when the objective is to predict the time until a cardiovascular event rather than simply classify patients as high- or low-risk [12].

The evaluation of CVD prediction models requires more than reporting accuracy alone. CVD datasets can contain substantial class imbalance, meaning that a model may achieve high accuracy while failing to identify patients who actually have elevated cardiovascular risk [13], [15]. Therefore, studies commonly evaluate models using sensitivity, specificity, precision, F1-score, and ROC-AUC [13], [15]. For time-to-event prediction, the C-index is commonly used to evaluate discrimination [12]. Calibration is also important because a clinically useful risk model should produce predicted probabilities that correspond closely to observed event rates [12], [15]. Alongside predictive performance, explainability has become increasingly important because highly complex models can operate as “black boxes”. Techniques such as SHAP, LIME, permutation-based feature importance, and rule extraction can provide insight into the features influencing predictions [12], [13], [16]. This can improve transparency and potentially increase clinical confidence in ML-based decision-support systems.

Despite promising results, significant limitations remain in the existing literature. Small datasets can increase overfitting and limit the representativeness of models, while class imbalance can cause models to favour the majority low-risk class [12], [13]. The literature also demonstrates a high risk of methodological bias, particularly due to inadequate handling of missing data, inappropriate preprocessing, and overfitting [12], [15]. More importantly, external validation remains uncommon. Only six of 33 studies in one systematic review performed external validation, while another meta-analysis reported external validation in only 14.29% of studies [12], [15]. This limits confidence that models will generalise to independent populations. Geographic and demographic representation is another concern, as many EHR-based studies originate from high-income Western countries, potentially limiting applicability to underrepresented populations [11], [15]. Finally, inconsistent reporting of calibration, decision thresholds, and evaluation procedures makes direct comparison between models difficult [12], [15].

Overall, the literature demonstrates that ML and DL offer considerable potential for improving CVD risk prediction, particularly through their ability to model complex relationships within clinical data. However, predictive performance alone is insufficient for clinical adoption. Robust preprocessing, appropriate handling of class imbalance, comprehensive evaluation, external validation, generalisability, and explainability remain important requirements. These limitations provide a basis for further investigation into reliable and interpretable ML approaches for cardiovascular disease risk prediction.

3. Methodology

3.1 Dataset

The cardiovascular disease dataset used is publicly available on Kaggle [17] and contains approximately 70,000 patient records with 13 variables in CSV format. The dataset contains demographic, physiological, and lifestyle-related attributes. It is designed as a binary classification problem, where the target variable indicates whether a patient has been diagnosed with cardiovascular disease (cardio = 1) or not (cardio = 0).

Table 1. Dataset feature descriptions

Feature	Description
ID	Patient's assigned number
Age	Patient's age in days (can be converted to years).
Height	Patient's height in centimetres.
Weight	Patient's weight in kilograms.
Gender	Patient's gender (1 = female, 2 = male). (later mapped to 0/1).
Systolic BP	Systolic blood pressure measurement.
Diastolic BP	Diastolic blood pressure measurement.
Cholesterol	Cholesterol level (1 = normal, 2 = above normal, 3 = well above normal).
Glucose	Glucose level (1 = normal, 2 = above normal, 3 = well above normal).
Smoking	Whether the patient smokes (0 = no, 1 = yes).
Alcohol	Whether the patient consumes alcohol (0 = no, 1 = yes).
Physical Activity	Whether the patient engages in physical activity (0 = no, 1 = yes).
Cardio	Presence of cardiovascular disease (0 = no, 1 = yes).

3.2 Preprocessing

Preprocessing was performed to ensure data quality, consistency and suitability for machine learning models.

Missing Values: Analysis revealed no missing values across the dataset. Therefore, imputation strategies (median for numerical, mode for categorical) were prepared but not applied.

Duplicate Removal: No duplicate rows were identified so no records were removed in this step.

Outlier Detection: The original dataset contained several physiologically implausible measurements, particularly for blood pressure, height, and weight. Range-based filtering was therefore applied to remove clearly unrealistic observations. Systolic blood pressure (ap_hi) was restricted to 80-300 mmHg, diastolic blood pressure (ap_lo) to 40–200 mmHg, height to 120–220 cm, and weight to 30–180 kg. Records where systolic pressure was lower than

diastolic pressure were also removed because they were considered physiologically inconsistent. Using these criteria, 1,390 records were removed, reducing the dataset from 70,000 to 68,610 observations.

Data Cleaning: Invalid or inconsistent values were removed.

Feature Scaling: Feature scaling was not applied to the complete dataset during the initial preprocessing stage. Scaling was instead incorporated into the modelling pipeline where required. In particular, numerical features were standardised for the Support Vector Machine because SVM performance is sensitive to differences in feature magnitude. Scaling was not required for the Decision Tree model. To prevent data leakage, the scaler was fitted using only the training data and subsequently applied to validation and test data.

Encoding: Gender mapped to binary (0 = female, 1 = male) for consistency with binary representations. Cholesterol and glucose were retained as ordinal variables because their values represent ordered severity levels. Lifestyle variables (smoke, alco, active) already binary and left unchanged.

3.3 Exploratory Data Analysis

Descriptive statistics

The average age of participants is approximately 53 years, with a range roughly between 29 and 65 years old. The average height is 164.4 cm, and the average weight is 74.1 kg. Blood pressure readings show a mean systolic of 126.7 mmHg and a mean diastolic of 81.3 mmHg. Categorical data indicates that the majority of the population is active (80.3%), while smoking (8.8%) and alcohol consumption (5.3%) are less prevalent.

Correlation analysis and Heatmap

The correlation analysis evaluates the linear relationship between various biometric features and the presence of cardiovascular disease. The features were ranked by their correlation coefficients:

Strongest Predictors:

Systolic blood pressure shows the highest positive correlation with cardiovascular disease at 0.428, followed by diastolic blood pressure at 0.340. This suggests that as blood pressure increases, the likelihood of having CVD significantly increases.

Moderate Predictors:

Age (0.239), cholesterol (0.221), and weight (0.180) also exhibit notable positive correlations with the target variable.

Weak Positive Predictors:

Glucose levels (0.090), and gender (0.007) have very low positive correlations, suggesting they have minimal direct linear impact on the CVD status in this dataset.

Negative Correlations:

Several features show a slight negative correlation. Physical activity at -0.038, which suggests a minor protective effect against CVD. Other features with very weak negative correlations include smoking (-0.016), height (-0.012), and alcohol consumption (-0.008).

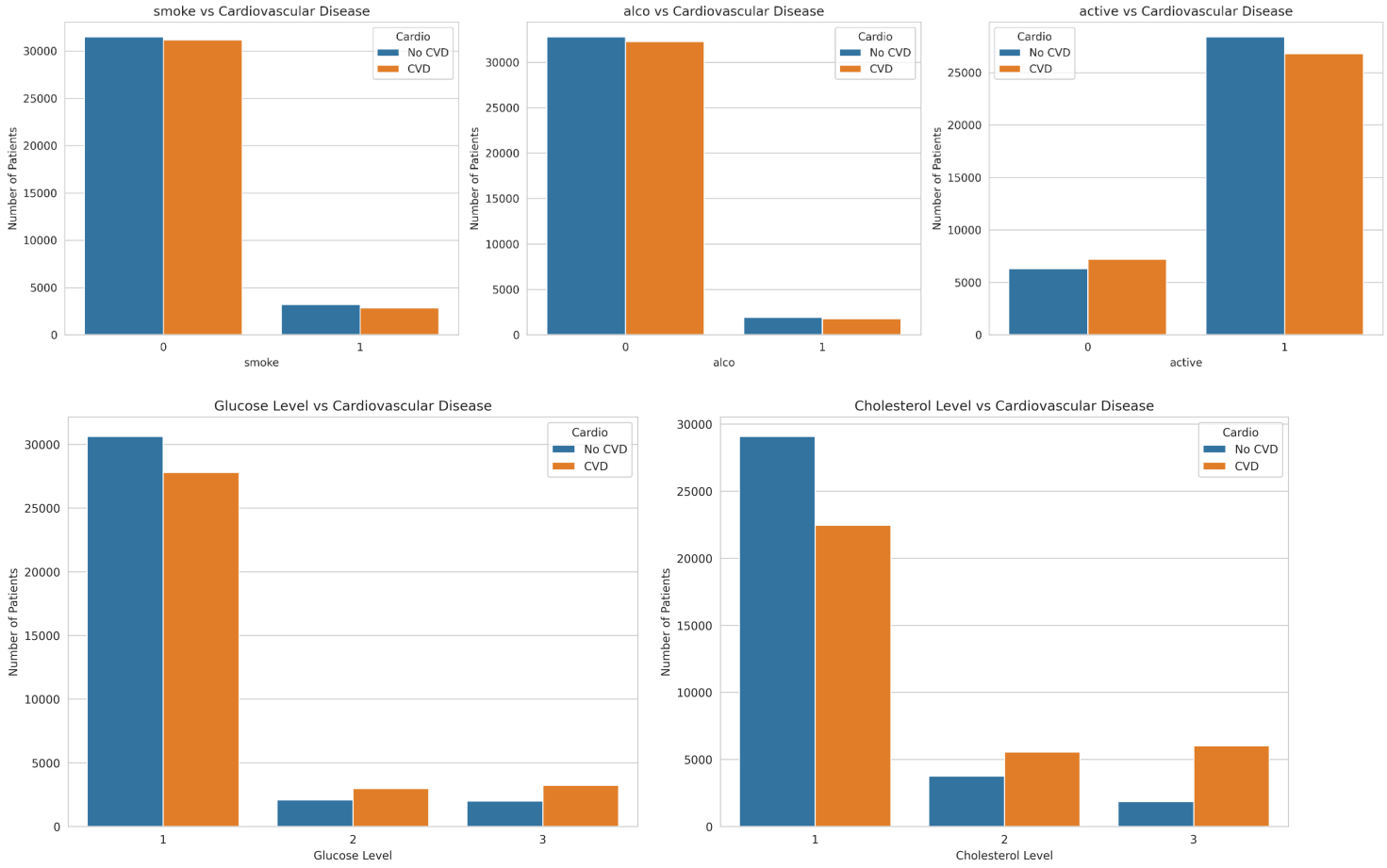


Figure 1. Distribution of categorical variables (smoking, alcohol, physical activity, glucose, cholesterol) by cardiovascular disease status

The Correlation Heatmap of Cardiovascular Disease Features:

It provides a visual representation of these relationships, including correlations between independent variables:

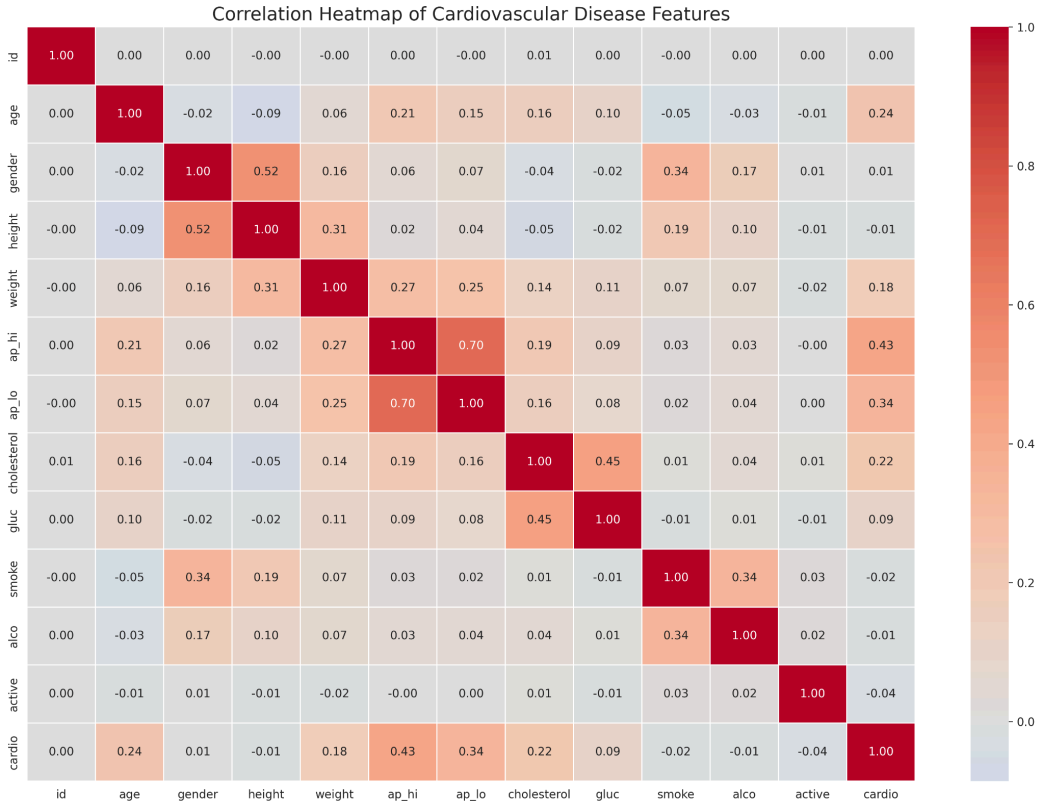


Figure 2. Correlation heatmap of cardiovascular disease features

Histograms

The distribution of numerical features gives us insights into the population's health profile:

1. Age and Height:

Age shows a slightly left skewed distribution, and heights follow a near normal distribution centred around 160cm to 170cm

2. Weight:

This feature is right skewed, telling us that the portion of the population has significantly higher body mass

3. Blood Pressure:

Both systolic and diastolic pressures show sharp peaks at common clinical values like 120/80mmHg

4. Categorical Features:

Distribution plots show that most patients have normal levels of cholesterol and glucose, are non-smokers, do not consume alcohol and are physically active.

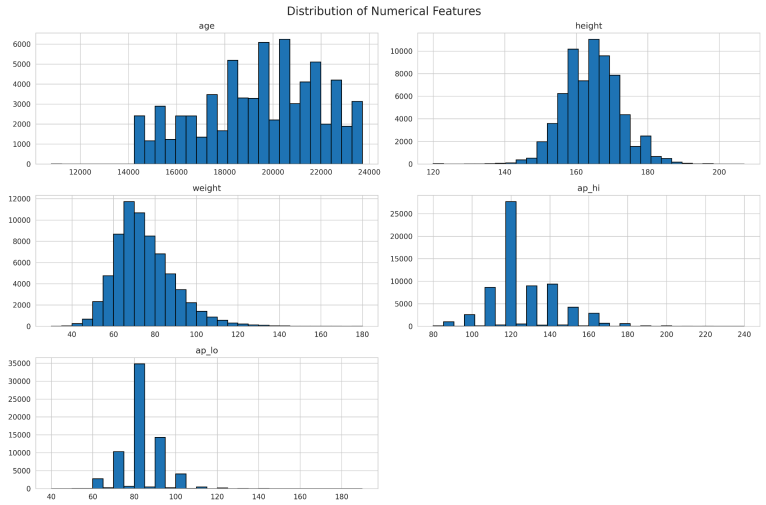


Figure 3. Distribution of numerical features (age, height, weight, systolic and diastolic blood pressure)

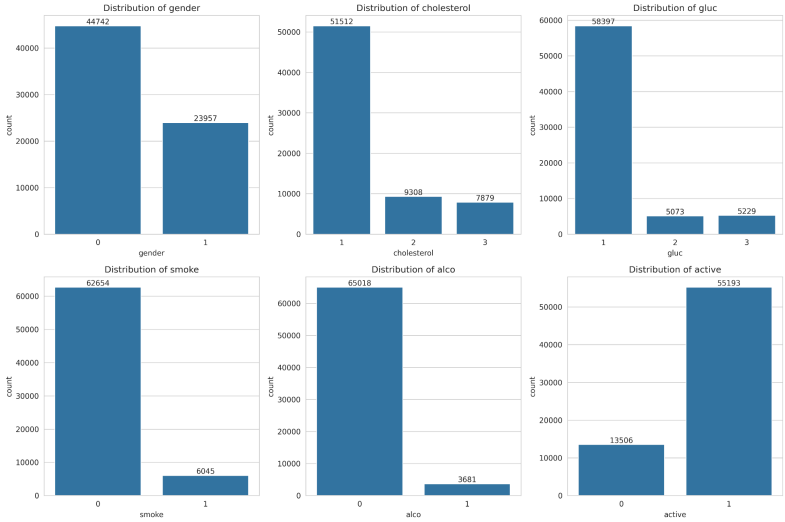


Figure 4. Distribution of categorical features (gender, cholesterol, glucose, smoking, alcohol, physical activity)

Boxplots

Boxplots reveal significant differences in biometric indicators between the No CVD and CVD groups:

1. Blood Pressure:

There is a marked increase in both systolic and diastolic blood pressure for patients with CVD compared to those without.

2. Age and Weight:

Patients with CVD tend to be older and heavier.

3. Height:

This feature shows very little variation between the two groups, suggesting it is not a strong indicator of cardiovascular health status.

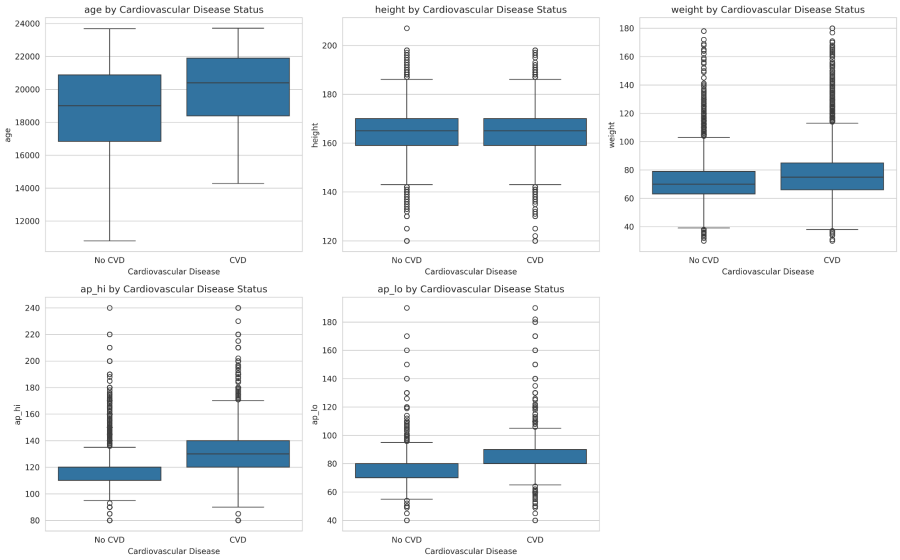


Figure 5. Boxplots of numerical features by cardiovascular disease status

Scatter plots and Pairwise Relationships

1. Systolic vs Diastolic Blood Pressure:

A clear positive correlation exists between these two features. A high density of patients with CVD is observed at higher blood pressure levels (above 140/90 mmHg).

2. Height vs Weight:

While there is a general correlation between height and weight, the CVD status is more prevalent among individuals with higher weights across all height ranges. This tells us that height is not a strong indicator of CVD.

3. Feature Pairs:

Pairwise plots further illustrate that age and weight have visible shifts when coloured by CVD status, whereas height remains relatively stagnant.

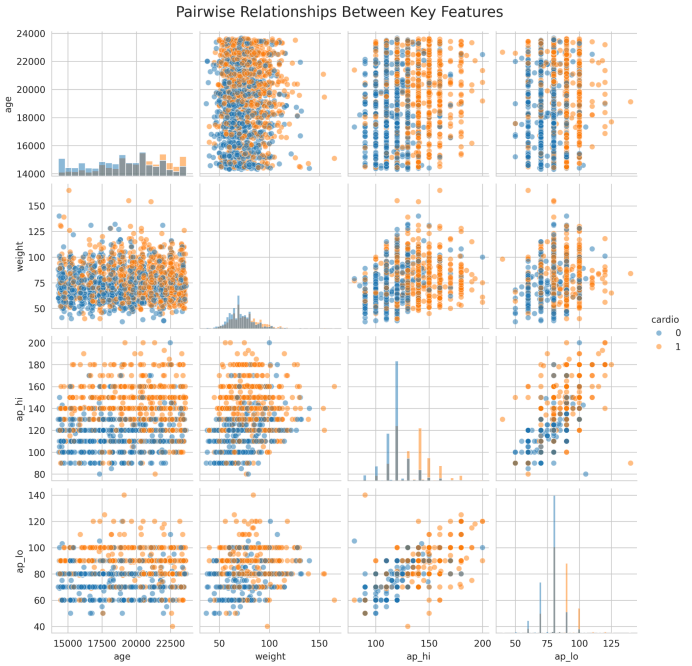


Figure 6. Pairwise relationships between key numerical features, coloured by cardiovascular disease status

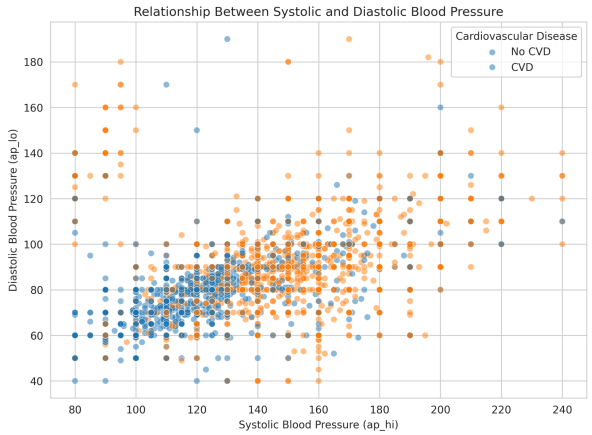


Figure 7. Relationship between systolic and diastolic blood pressure

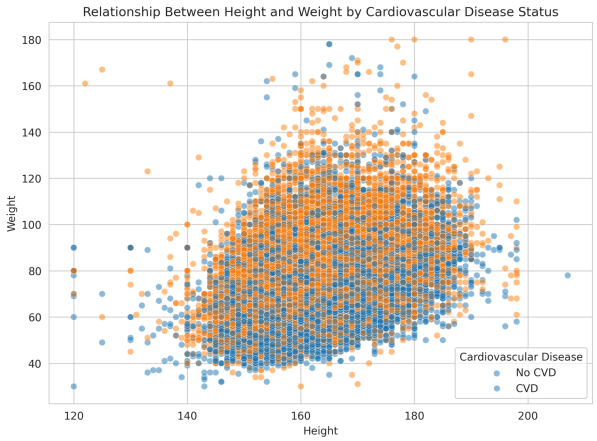


Figure 8. Relationship between height and weight by cardiovascular disease status

3.4 Feature Engineering

Feature engineering was performed to derive additional variables from the existing demographic, physiological, and lifestyle features. The purpose of this process was to create new representations of the available data that may provide additional information for predicting cardiovascular disease.

Table 2. Engineered features, rationale, construction method, and expected benefits

Engineered Feature	Why Was It Created?	How Was It Created?	Expected Benefits
Age In Years	Original age was in days	Age ÷ 365.25	Improves interpretability and supports age based analysis
BMI	Combines height and weight	Weight / Height²	Provides combined representation of body size relative to height
Pulse Pressure	Difference between systolic and diastolic BP	ap_hi - ap_lo	May capture additional blood pressure characteristics
Mean Arterial Pressure (MAP)	Combines BP measurements	Ap_lo + (Pulse Pressure)/3	Provides average representation of arterial pressure
BMI Category	Represents broad BMI groups	Divided into four categories	May capture differences across weight categories
Age Category	Represents broad age groups	Divided into defined ranges	May capture non-linear age-related patterns
Hypertension Indicator	Combines systolic and diastolic BP thresholds	ap_hi ≧ 140 or ap_lo ≧ 90	Provides representation of elevated blood pressure
Lifestyle Risk Score	Combines included lifestyle-related risk factors	Smoking + alcohol + inactivity	Represents cumulative presence of lifestyle risk factors

The final dataset contains 68,610 observations and 20 variables. Eight new features were added to the original dataset. Validation checks showed that all eight engineered features were successfully created, no missing values were present in the engineered features, no infinite values were introduced, the blood pressure consistency check removed records where systolic blood pressure was lower than diastolic blood pressure before blood-pressure-derived features were created, and all original and engineered features were retained for the next stage.

The full set of features was then passed to the feature selection stage (Section 3.5), where the contribution of the original and engineered variables to cardiovascular disease prediction was evaluated.

3.5 Feature Selection

3.5.1 Multicollinearity Check

The engineered feature set was screened for multicollinearity before formal selection, since several features were deliberately derived from the same underlying variables. **age** and **age_years** were found to be perfectly correlated (r = 1.00), and **mean_arterial_pressure** correlated strongly with both **ap_lo** (r = 0.94) and **ap_hi** (r = 0.92), confirming these are largely redundant with variables already in the dataset.

The Variance Inflation Factor (VIF) was then computed, after removing exact algebraic duplicates (**age**, **pulse_pressure**, **mean_arterial_pressure**, **lifestyle_risk_score**, each a deterministic linear function of columns already present):

Table 3. Variance Inflation Factor (VIF) scores for candidate features

Feature	VIF
weight	88.14
bmi	78.29
height	24.04
age_years	6.08
age_category	6.00

All remaining features had VIF below 5, indicating no meaningful multicollinearity concern. **weight** and **height** were dropped in favour of **bmi**, which is purpose-built to combine them. **age_category** and **bmi_category** were dropped in favour of their continuous counterparts, since a Decision Tree can learn its own splits without losing information to manual binning.

3.5.2 Formal Feature Selection Method

On the resulting 11-feature candidate set, three ranking methods were applied to the training data: ANOVA F-score, Mutual Information, and Decision Tree feature importance.

Table 4. Feature ranking by ANOVA F-score, Mutual Information, and Decision Tree Importance

Feature	ANOVA F	ANOVA p	Mutual Info	Tree Importance
ap_hi	12213.57	<0.001	0.118	0.766
age_years	3364.76	<0.001	0.033	0.125
hypertension	10976.85	<0.001	0.089	0.003
ap_lo	7045.35	<0.001	0.071	0.001
cholesterol	2794.44	<0.001	0.027	0.073
bmi	2028.35	<0.001	0.028	0.022
gluc	469.35	<0.001	0.008	0.007
active	82.60	<0.001	0.006	0.001
smoke	13.33	<0.001	0.001	0.001
gender	4.74	0.029	0.004	<0.001
alco	1.20	0.274	0.000	0.000

alco failed the significance test outright and returned zero Mutual Information and zero Tree importance across all three methods, a consistent null result, so it was excluded. This may reflect under-reporting bias in self-reported alcohol use (present in only 5.3% of the dataset). **gender** was a borderline case ($p = 0.029$ but near-zero effect size); with ~55,000 training rows even trivial effects reach significance, so the p-value was not used to judge its relevance. It was retained as a standard clinical covariate that adds little noise.

3.5.3 Final Selected Feature Set

Based on the screening and ranking above, the final selected feature set is: **age_years, gender, ap_hi, ap_lo, cholesterol, gluc, smoke, active, bmi, hypertension** (10 features), reduced from the full 19-column engineered dataset.

3.5.4 Empirical Validation: Comparing Feature Sets

A baseline Decision Tree (max_depth = 6, 5-fold CV) was compared across the original, full engineered, and selected feature sets to confirm the reduction did not cost predictive performance:

Table 5. Cross-validation accuracy and F1-score across the original, full engineered, and selected feature sets

Feature Set	Number of Features	CV Accuracy	CV F1 Score
Original	11	0.7274 ± 0.0053	0.7096 ± 0.0091
Full Engineered	19	0.7279 ± 0.0049	0.7117 ± 0.0068
Selected	10	0.7278 ± 0.0054	0.7142 ± 0.0071

The selected set matches the full engineered set on accuracy and slightly exceeds it on F1, using close to half as many features, confirming the redundant features removed were not contributing real predictive value.

3.6 Model Development

Two classical machine learning algorithms were selected as base classifiers for this project: a Decision Tree and a Support Vector Machine (SVM). These were chosen to represent two structurally different approaches to classification, a rule-based, non-parametric model that partitions the feature space recursively, and a margin-based model that separated classes using a kernel-transformed decision boundary. This structural difference was expected to produce complementary error patterns between the two models, which subsequently motivated the development of an ensemble model (Section 3.9).

Both models were implemented using scikit-learn. The Decision Tree was trained directly on the 10 selected features (Section 3.5.3), as tree-based models are invariant to feature scaling. The SVM was trained on the same feature set after standardisation (Section 3.2), since SVM performance is sensitive to differences in feature magnitude. Both models followed an identical development protocol: an 80/20 train-test split was performed first (Section 4), after which each model underwent 5-fold cross-validation on the training set to select its best hyperparameter configuration (Section 3.7), before being retrained on the full training set and evaluated once on the held-out test set (Section 5).

3.7 Hyperparameter Optimisation

Table 6. Decision Tree cross-validation performance by max depth

Max Depth	CV Accuracy	CV F1
5	0.7266 ± 0.0046	0.7035 ± 0.0101
10	0.7239 ± 0.0052	0.7084 ± 0.0060
15	0.7057 ± 0.0048	0.6895 ± 0.0060
20	0.6817 ± 0.0052	0.6666 ± 0.0075

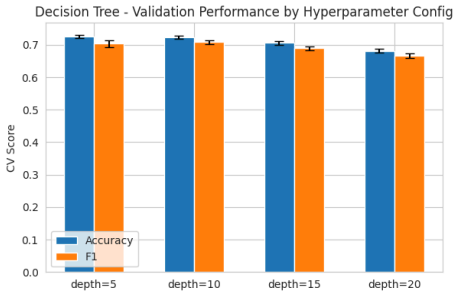


Figure 9. DT validation performance by hyperparameter configuration

Accuracy declines as depth increases, indicating overfitting. **max_depth=5** was selected on mean CV accuracy, though **max_depth=10** had marginally higher F1 (0.7084 vs 0.7035); accuracy was used as the primary criterion given the near-balanced classes (50.5%/49.5%). Table 7. SVM cross-validation performance by kernel

Kernel	CV Accuracy	CV F1
linear	0.6953 ± 0.0023	0.6359 ± 0.0030
poly	0.7192 ± 0.0033	0.6823 ± 0.0040
rbf	0.7273 ± 0.0049	0.7004 ± 0.0056
sigmoid	0.6440 ± 0.0091	0.6398 ± 0.0083

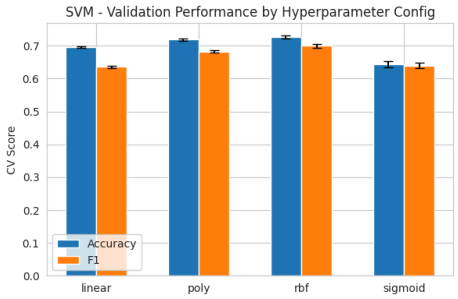


Figure 10. SVM validation performance by hyperparameter configuration

RBF outperformed the alternatives clearly and was selected.

3.8 Cross-validation Strategy

A consistent cross-validation strategy was applied throughout the project to ensure all model comparisons were performed on identical data splits. Stratified 5-fold cross-validation (scikit-learn's **StratifiedKFold**) was used for every hyperparameter search and model comparison, preserving the approximately 50.5% / 49.5% class balance (Section 3.1) within each fold. A fixed random seed (**random_state=42**) was used consistently when generating these folds, ensuring that the Decision Tree hyperparameter search, the SVM hyperparameter search, the ensemble configuration search (Section 3.9), and the final Decision Tree-SVM-Ensemble comparison used for statistical testing (Section 3.11) all drew from the same five fold splits. This consistency is a methodological requirement rather than a stylistic choice: the paired t-test and Wilcoxon signed-rank test used in Section 3.11 compare performance across matching folds between models, so using different or independently re-shuffled folds for each model would invalidate that pairing.

3.9 Ensemble Learning

Given the complementary error profile observed between the Decision Tree and SVM (Section 5.1), an ensemble model was developed to test whether combining the two would yield a statistically meaningful improvement over either model individually. Rather than implementing a single ensemble configuration, four candidate configurations were evaluated using the same 5-fold stratified cross-validation protocol as the base models, to determine empirically which combination strategy performed best before committing to one for final testing.

The four configurations tested were: (1) soft voting with equal weighting between the Decision Tree and SVM, (2) soft voting weighted 2:1 in favour of the Decision Tree, (3) soft voting weighted 2:1 in favour of the SVM, and (4) a stacking classifier using a logistic regression meta-learner trained on the base models' outputs. Soft voting combines predictions by averaging each base model's predicted class probabilities, whereas stacking trains a second-level model to learn how to combine the base models' outputs.

A methodological note on the SVM component: The final SVM evaluated in Section 5.1 was trained with **probability=False**, using the raw decision function for ROC-AUC calculation. Soft voting requires calibrated class probabilities from each base estimator in order to average them, so the SVM used within the ensemble was configured with **probability=True**, which enables Platt scaling on top of the same underlying model (identified kernel and hyperparameter as selected in Section 3.7). This is not a hyperparameter change and does not violate the protocol against re-tuning after test-set evaluation; it is a technical requirement for producing calibrated probability outputs rather than a decision score.

All configurations were evaluated using identical 5-fold cross-validation splits on the training set:

Table 8. Cross validation performance (5-fold, mean ± standard deviation) of the four ensemble models tested

Configuration	CV Accuracy	CV F1
Equal Soft Voting	0.7303 ± 0.0055	0.7056 ± 0.0073
Soft Voting (DT-weighted 2:1)	0.7300 ± 0.0049	0.7051 ± 0.0076
Soft Voting (SVM-weighted 2:1)	0.7295 ± 0.0054	0.7052 ± 0.0063
Stacking (logistic meta-learner)	0.7299 ± 0.0042	0.7071 ± 0.0076

Soft voting with equal weighting achieved the highest mean cross-validation accuracy (0.7303 ± 0.0055) and was selected as the final ensemble configuration. Notably, the weighted variants did not outperform equal weighting, suggesting the two models contribute comparably useful information to the vote, consistent with their near-identical accuracy scores. The stacking classifier achieved the highest mean F1-score (0.7071 ± 0.0076) but was not selected, as accuracy was used as the primary selection criterion given the near-balanced class distribution (50.5% / 49.5%), consistent with the model selection criterion used for the individual models in Section 3.7.

3.10 Explainable AI

SHAP (SHapley Additive exPlanations) was used to provide global and local explanations for the Decision Tree, SVM, and Ensemble models. Given the computational cost of model-agnostic SHAP, particularly for the SVM and Ensemble, values were computed using **shap.Explainer** with the permutation algorithm, using a background sample of 100 training observations and explaining 300 held-out test observations. The same 300 observations were used across all three models to ensure their explanations were directly comparable. Because the Decision Tree and Ensemble output predicted class probabilities while the final SVM (trained with **probability=False**, Section 3.9) outputs a raw decision-function margin, SHAP values were computed against each model's respective native output. To allow fair comparison of relative feature importance despite this difference in output scale, each model's SHAP values were normalised to a percentage of that model's total mean absolute SHAP importance, rather than compared as raw magnitudes.

The global SHAP analysis identified blood-pressure-related variables, age and cholesterol as the principal factors influencing CVD (cardiovascular disease) predictions. Although there was considerable agreement between the models

regarding the most influential features, differences were observed in the relative importance assigned to individual variables.

For the Decision Tree, systolic blood pressure (ap_hi) was the dominant feature, accounting for 63.87% of normalised mean absolute SHAP importance. This was followed by age (18.94%), cholesterol (11.58%), hypertension (2.10%) and glucose (1.46%). The concentration of importance in ap_hi indicates that the Decision Tree relied heavily on systolic blood pressure when separating patients with and without CVD.

The SVM distributed its feature importance more broadly. Hypertension was ranked first at 29.36%, closely followed by systolic blood pressure at 27.20%. Age accounted for 13.66%, cholesterol for 11.90% and BMI for 5.82%. In contrast with the Decision Tree, the SVM therefore placed considerably greater relative importance on the engineered hypertension feature.

The Ensemble combined characteristics of both individual models. Systolic blood pressure remained the most influential feature at 44.94%, followed by hypertension (16.87%), age (16.30%), cholesterol (11.54%) and BMI (3.40%). The Ensemble therefore retained the Decision Tree's strong emphasis on systolic blood pressure while incorporating the greater importance assigned to hypertension by the SVM.

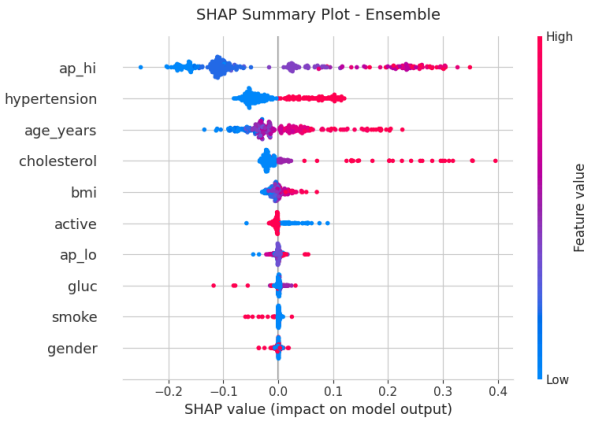


Figure 11. SHAP summary plot, Ensemble

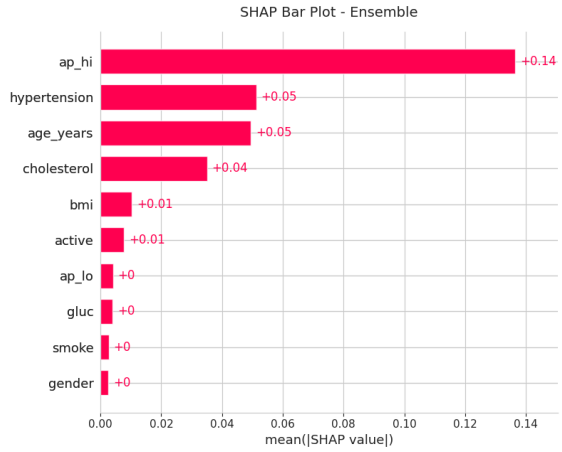


Figure 12. SHAP bar plot, Ensemble

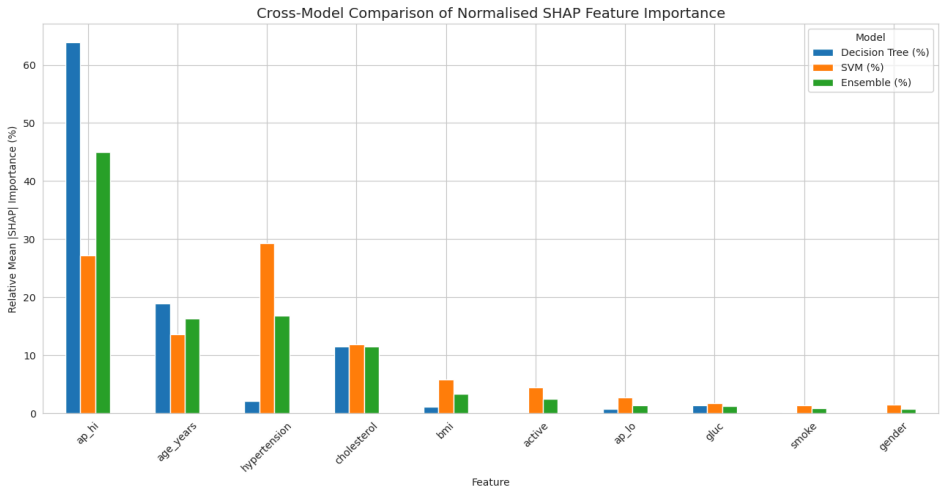


Figure 13. Cross-model comparison of normalised SHAP feature importance (Decision Tree, SVM, Ensemble)

The Ensemble SHAP bar plot summarises global feature importance using mean absolute SHAP values. Systolic blood pressure (ap_hi) was the most influential feature, followed by hypertension, age, cholesterol, and BMI, consistent with the ordering observed in the Ensemble SHAP summary plot.

Across all three models, systolic blood pressure, age, hypertension and cholesterol consistently appeared among the most influential predictors. Systolic blood pressure had the highest average relative importance across the three models (45.33%). However, hypertension was engineered from systolic and diastolic blood pressure and should therefore not be interpreted as an entirely independent clinical signal. Rather, the combined importance of ap_hi and hypertension indicates that blood-pressure-related information played a substantial role in the fitted models.

The SHAP summary plots also provided information about the direction of these effects. Higher systolic blood pressure values generally produced positive SHAP contributions, pushing predictions towards the CVD class, whereas lower systolic blood pressure values generally produced negative contributions. Older age similarly tended to increase the model output towards CVD, while higher cholesterol values could produce substantial positive contributions. These patterns were particularly evident in the Decision Tree and Ensemble summary plots.

The Ensemble SHAP dependence plot for systolic blood pressure demonstrated a clear nonlinear relationship between ap_hi and the model output. Lower systolic blood pressure values, particularly those around 90-120 mmHg in the explained sample, generally produced negative SHAP values and therefore reduced the model's predicted probability of CVD. SHAP contributions approached zero around the intermediate range and became increasingly positive at higher systolic blood pressure values, particularly from approximately 130-140 mmHg onwards.

Cholesterol was automatically selected by SHAP as the interaction-colouring feature. Variation in SHAP values among observations with similar systolic blood pressure suggests that the contribution assigned to blood pressure by the fitted Ensemble may vary according to other patient characteristics. However, this visualisation represents model behaviour and should not be interpreted as evidence of a causal clinical interaction.

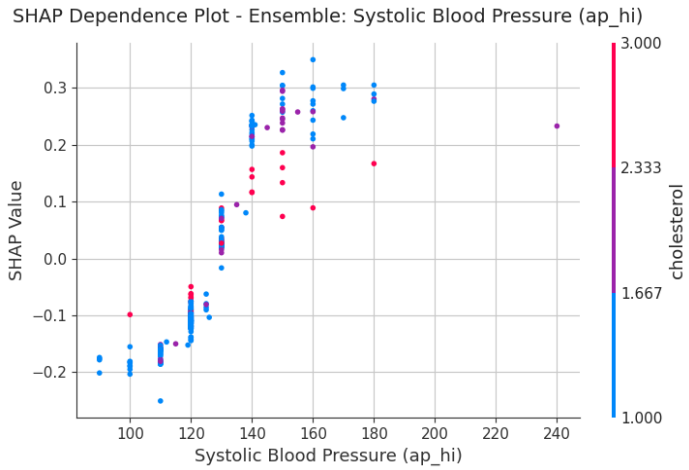


Figure 14. SHAP dependence plot, Ensemble: systolic blood pressure (ap_hi)

Local SHAP explanations were used to investigate how individual feature values contributed to specific Ensemble predictions. For the selected correctly classified CVD patient, the Ensemble predicted a CVD probability of 0.8667 compared with a SHAP baseline output of approximately 0.497. The patient was approximately 60.2 years old, had systolic blood pressure of 140 mmHg, diastolic blood pressure of 80 mmHg, cholesterol level 2, BMI of approximately 24.03, and a positive hypertension indicator.

Systolic blood pressure produced the largest positive contribution to this patient's prediction, increasing the model output by approximately +0.23. Age contributed approximately +0.09 and the hypertension indicator approximately +0.05. The remaining features had relatively small effects. Therefore, the elevated prediction for this patient was primarily associated with the combined influence of systolic blood pressure, age, and hypertension.

In contrast, the selected correctly classified No-CVD patient received a CVD probability of 0.1340. This patient was approximately 46.0 years old, had systolic blood pressure of 110 mmHg, diastolic blood pressure of 90 mmHg, cholesterol level 1, and BMI of approximately 25.78. Although the patient had a positive hypertension indicator, systolic blood pressure of 110 mmHg produced the strongest negative SHAP contribution at approximately −0.25. Younger age contributed approximately −0.07, while cholesterol and diastolic blood pressure also produced smaller negative contributions. The positive contribution associated with hypertension, approximately +0.03, was therefore outweighed by the combined negative contributions of the other features, resulting in the substantially lower CVD probability.

This comparison demonstrates the value of local explainability. The presence of a single risk-related feature did not determine the prediction independently; instead, the final output reflected the combined contributions of the patient's complete feature profile.

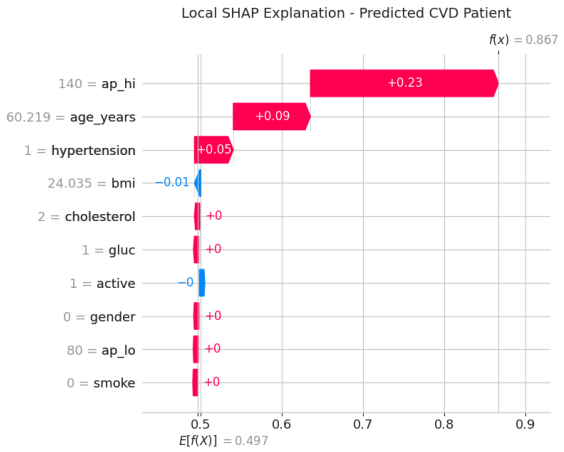


Figure 15. Local SHAP explanation for a correctly classified CVD patient, Ensemble

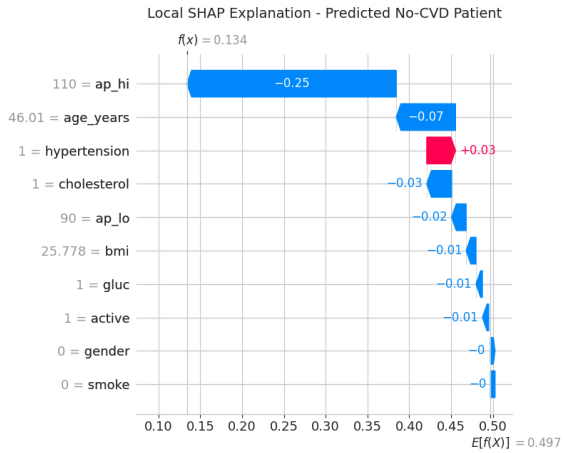


Figure 16. Local SHAP explanation for a correctly classified No-CVD patient, Ensemble

3.11 Statistical Analysis

To determine whether performance differences between models were statistically meaningful rather than due to chance, paired statistical tests were conducted on the per-fold cross-validation scores from the same five stratified folds used throughout the project (Section 3.8). For each comparison, the null hypothesis (H_0) stated that there was no difference in mean performance between the two models, against the alternative hypothesis (H_1) that a difference existed. A significance level of $\alpha = 0.05$ was used throughout.

The Shapiro-Wilk test was first applied to the paired fold-level differences to assess normality. Where normality was not rejected, a paired t-test was used; otherwise, the Wilcoxon signed-rank test would have been applied instead. Two required comparisons were conducted, Decision Tree vs Ensemble and SVM vs Ensemble, on both accuracy and F1-score, with the Decision Tree vs SVM comparison included for completeness. The results of these tests are presented and interpreted in Section 5.6. The per-fold scores are shown in Figures 17 and 18.

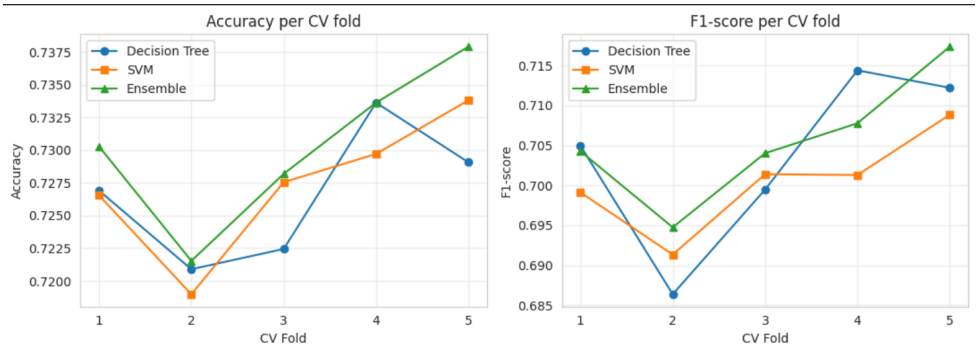


Figure 17. Accuracy and F1-score per cross-validation fold for the Decision Tree, SVM, and Ensemble

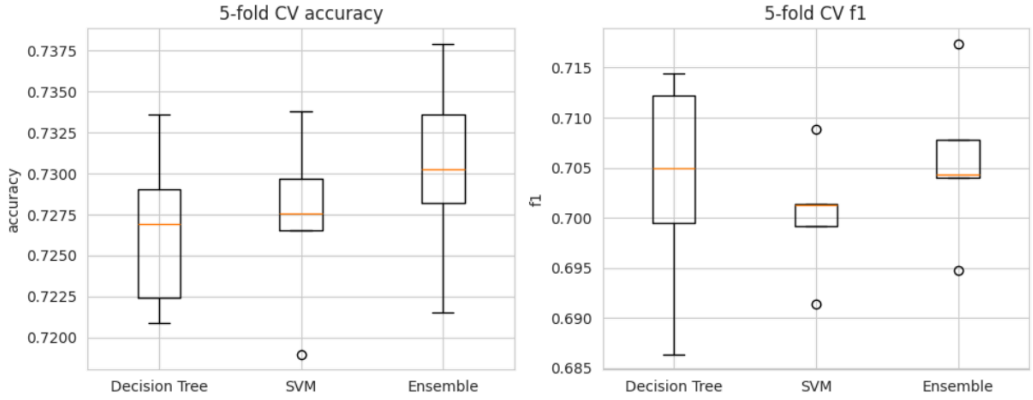


Figure 18. Distribution of 5-fold cross-validation accuracy and F1-score by model

4. Experimental Setup

Environment. All experiments were implemented in Python in Google Colab using scikit-learn, NumPy, pandas, Matplotlib, SciPy, and SHAP. A fixed random seed (42) was used for the data split, the cross-validation folds, and all stochastic models, ensuring reproducible results.

Data partitioning and protocol. After preprocessing, feature engineering, and feature selection, the 68,610 records were split into an 80% training set (54,888 records) and a 20% held-out test set (13,722 records). The split was stratified. The test set was not used for tuning and was evaluated only once for each final model. All model selection was performed on the training set using stratified 5-fold cross-validation. The same fold splits were used for all models and configurations. This is what makes the fold scores directly comparable and allows paired statistical tests.

Hyperparameter configurations. Each individual model was evaluated with four configurations, and the best was chosen by the mean CV accuracy:

- Decision Tree: max_depth $\in$ {5, 10, 15, 20}. Depth controls model complexity, so this range covers strongly regularised trees through to nearly unrestricted ones. Selected: depth = 5.
- **SVM:** kernel $\in$ {linear, polynomial, RBF, sigmoid}. Features were standardised inside a pipeline, so the scaler was fitted only on each training fold, and there is no leakage into the validation fold. Other parameters were left at scikit-learn defaults. Selected: kernel = rbf.
- Ensemble: Models 1 and 2 were combined, each using its best configuration. Four ensemble configurations were compared: soft voting with equal weights, soft voting weighted towards the DT (2:1), soft voting weighted towards the SVM (1:2), and stacking with a logistic regression meta-learner (internal 5-fold CV).

The SVM was used with probability=True to produce probabilities for soft voting. Selected: Soft Voting (equal).

Evaluation metrics. Accuracy, precision, recall, specificity, F1-score, ROC-AUC, MCC, and Cohen's kappa on the test set. Cross-validation model selection and the statistical comparison used accuracy and F1.

5. Results and Discussion

5.1 Final Model Evaluation

Following hyperparameter optimisation, the Decision Tree with a maximum depth of 5 and the Support Vector Machine with an RBF kernel were selected as the individual models. Each model was retrained on the full training set using its selected configuration and evaluated once on the held-out test set, with no further tuning performed after the test set was used.

Table 9. Final test-set evaluation of the Decision Tree and SVM

Metric	Decision Tree (depth=5)	SVM (rbf)
Accuracy	0.7371	0.7366
Precision	0.7679	0.7737
Recall	0.6717	0.6608
Specificity	0.8012	0.8108
F1-score	0.7166	0.7128
ROC-AUC	0.7962	0.7904
MCC	0.4772	0.4773
Cohen's Kappa	0.4735	0.4723

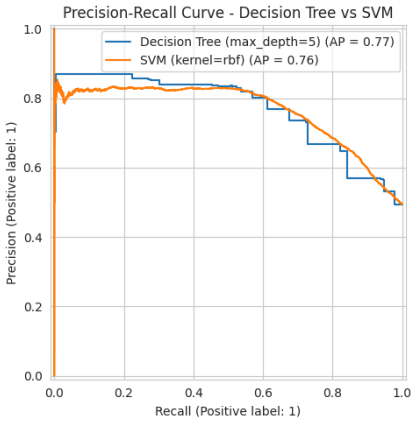


Figure 19. Precision-Recall curve, DT vs. SVM

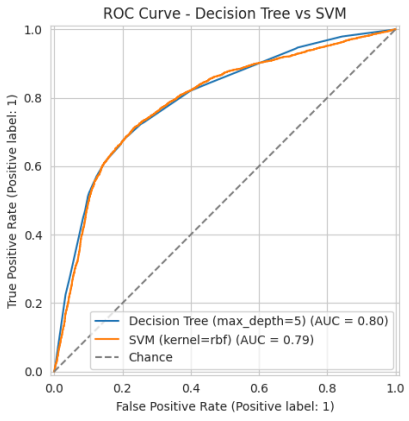


Figure 20. ROC curve, DT vs. SVM

The two models achieved very similar overall performance, with test accuracies of 73.71% for the Decision Tree and 73.66% for the SVM. Their MCC values were also almost identical at 0.4772 and 0.4773, respectively. However, the models displayed complementary strengths. The Decision Tree achieved higher recall (67.17% compared with 66.08%), indicating that it identified a slightly larger proportion of individuals with cardiovascular disease. In contrast, the SVM achieved higher precision (77.37% compared with 76.79%) and specificity (81.08% compared with 80.12%), indicating fewer false-positive classifications and stronger identification of the negative class.

The ROC-AUC values were also similar, with the Decision Tree achieving 0.7962 and the SVM achieving 0.7904. Overall, these results demonstrate that both models provided useful predictive performance, while making somewhat different classification errors. This complementary behaviour provided the motivation for evaluating an ensemble model. **Figure 21** shows the feature importance of the final Decision Tree. Systolic blood pressure (ap_hi) dominated by a wide margin, followed by age_years and cholesterol, consistent with the ranking in **Table 4** and the SHAP results in **Section 5.7**.

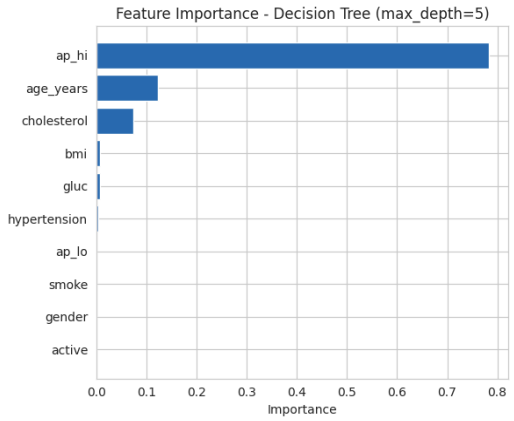


Figure 21. Feature importance of the final Decision Tree (max_depth = 5)

5.2 Effect of Feature Engineering

Feature engineering was evaluated by comparing model performance using the original features, the complete engineered feature set, and the final feature-selected dataset. The original dataset contained 11 usable predictive features, while the engineered dataset incorporated additional representations of age, body composition, blood pressure, hypertension, and lifestyle risk.

The results in **Table 5** show that feature engineering produced a small improvement in predictive performance. Mean cross-validation accuracy increased from 0.7274 using the original features to 0.7279 using the full engineered feature set, while the mean F1-score increased from 0.7096 to 0.7117. Although these improvements were modest, they indicate that transforming the raw variables into clinically meaningful representations provided additional predictive information.

The final selected feature set achieved almost the same accuracy as the full engineered feature set while achieving a slightly higher F1-score of 0.7142. This suggests that the additional engineered variables did not all provide unique predictive value once feature redundancy and statistical relevance were considered.

The results therefore support the use of feature engineering as a useful preprocessing step, although the relatively small performance differences indicate that feature engineering alone did not produce a substantial increase in predictive accuracy.

5.3 Effect of Feature Selection

Feature selection was subsequently applied to reduce redundancy and retain the variables that provided the greatest predictive value. Correlation and variance inflation factor analysis identified substantial redundancy between several variables, particularly between age and age in years, blood-pressure variables and mean arterial pressure, and height, weight, and BMI.

The selected feature set shown in **Table 5** contained 10 variables: age in years, gender, systolic blood pressure, diastolic blood pressure, cholesterol, glucose, smoking status, physical activity, BMI, and hypertension status. Despite using approximately half the number of variables in the full engineered dataset, the selected feature set maintained almost identical accuracy and achieved the highest F1-score.

This indicates that removing redundant and weakly informative variables did not reduce predictive performance. Instead, it produced a more compact representation of the data while retaining the information required for classification. The result is particularly relevant for interpretability because a smaller feature set makes it easier to understand the factors contributing to model predictions.

5.4 Hyperparameter Optimisation and Model Performance

Four configurations were evaluated for each individual model using stratified five-fold cross-validation on the training data. For the Decision Tree, maximum depth was varied across 5, 10, 15, and 20. For the SVM, linear, polynomial, RBF, and sigmoid kernels were evaluated.

The Decision Tree with maximum depth 5 was selected because it achieved the highest cross-validation accuracy of 0.7266. Although depth 10 produced a slightly higher F1-score, its accuracy was lower. Increasing the tree depth beyond 10 resulted in progressively poorer performance, suggesting that greater complexity did not improve generalisation. (see **Table 6**)

The RBF kernel produced the highest cross-validation accuracy and F1-score among the four SVM configurations and was therefore selected as the final SVM configuration. Its nonlinear kernel allows the model to capture relationships between the selected cardiovascular risk factors that cannot necessarily be represented by a linear decision boundary. (see **Table 7**)

The hyperparameter results demonstrate that model configuration had a meaningful effect on predictive performance. The selected configurations were therefore carried forward to final test-set evaluation and ensemble construction.

5.5 Ensemble Model Performance

Four ensemble configurations were evaluated: equal-weight soft voting, Decision Tree-weighted soft voting, SVM-weighted soft voting, and stacking with logistic regression as the meta-learner.

The equal-weight soft voting ensemble achieved the highest cross-validation accuracy of 0.7303 and was therefore selected as the final ensemble. Although the stacking configuration produced the highest F1-score, the equal soft voting model was selected because accuracy was the primary model-selection criterion. (see **Table 8**)

The selected ensemble was retrained using the full training set and evaluated once on the held-out test set.

Table 10. Final test-set evaluation of the Decision Tree, SVM, and selected model

Metric	Decision Tree	SVM	Ensemble
Accuracy	0.7371	0.7366	0.7389
Precision	0.7679	0.7737	0.7747
Recall	0.6717	0.6608	0.6659
Specificity	0.8012	0.8108	0.8103
F1-score	0.7166	0.7128	0.7162
ROC-AUC	0.7962	0.7904	0.8015
MCC	0.4772	0.4773	0.4816
Cohen's Kappa	0.4735	0.4723	0.4770

The ensemble achieved the highest test accuracy at 0.7389, compared with 0.7371 for the Decision Tree and 0.7366 for the SVM. It also achieved the highest precision (0.7747), ROC-AUC (0.8015), MCC (0.4816), and Cohen's Kappa (0.4770). However, the ensemble did not outperform both individual models on every metric. Its recall of 0.6659 was between that of the Decision Tree and SVM, while its F1-score of 0.7162 was very close to the Decision Tree's 0.7166.

These results indicate that soft voting benefited from the complementary behaviour of the two base models, but did not simply combine the strongest metric from each model. Instead, it produced a balanced performance profile, with improvements in several overall discrimination and agreement measures.

Precision-Recall and ROC Curves

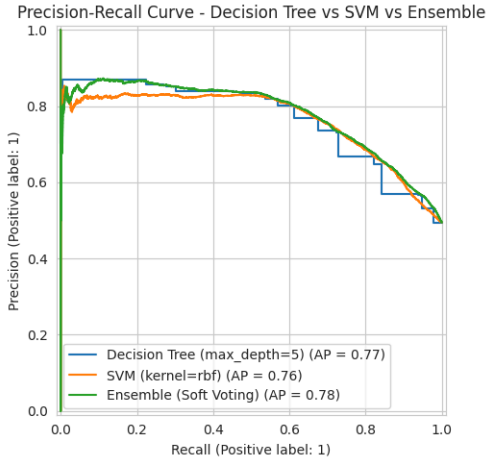


Figure 22. Precision-Recall curve, Decision Tree vs SVM vs Ensemble

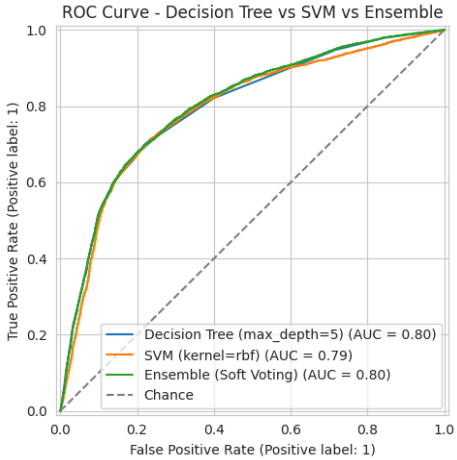


Figure 23. ROC curve, Decision Tree vs SVM vs Ensemble

The precision-recall curves further demonstrate the classification behaviour of the models. The ensemble generally maintains a favourable balance between precision and recall across classification thresholds. This is consistent with its test-set precision of 0.7747 and recall of 0.6659.

The ROC curves provide a visual comparison of the models across different classification thresholds. The ensemble achieved the highest ROC-AUC at 0.8015, compared with 0.7962 for the Decision Tree and 0.7904 for the SVM. The

ensemble curve is generally positioned above the two individual models across a range of false-positive rates, supporting the numerical AUC results.

Confusion Matrix

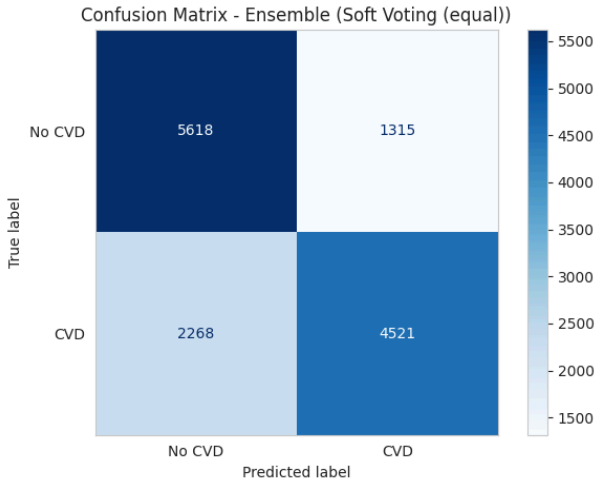


Figure 24. Confusion matrix for the Ensemble model on the held-out test set

The ensemble confusion matrix contained 5,618 true negatives, 1,315 false positives, 2,268 false negatives, and 4,521 true positives. The relatively larger number of false negatives compared with false positives is consistent with the ensemble's recall of 66.59%, indicating that a proportion of individuals with cardiovascular disease were still incorrectly classified as not having the disease.

Learning Curve

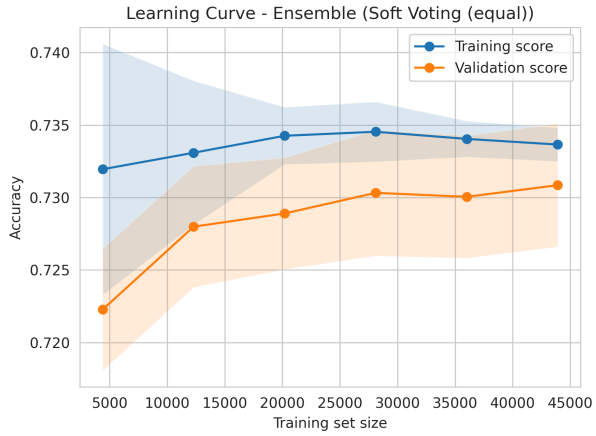


Figure 25. Learning curve for the Ensemble model

The learning curve shows that training and validation accuracy converge toward a relatively small gap, with approximately 0.734 training accuracy and 0.731 validation accuracy at the largest training size. Validation performance improves as additional training observations are introduced before beginning to plateau at approximately 28,000 observations. The relatively small and stable difference between training and validation performance provides no clear evidence of severe overfitting.

5.6 Statistical Comparison of the Models

To determine whether observed differences in cross-validation performance were statistically meaningful, paired statistical tests were conducted using the same five cross-validation folds for each model. Shapiro-Wilk tests were first applied to the paired differences. All resulting p-values were greater than 0.05, so the assumption of normally distributed paired differences was not rejected and paired t-tests were used.

Table 11. Paired statistical test results comparing the Decision Tree, SVM, and Ensemble

Comparison	Metric	t-statistic	Mean Difference	Shapiro-Wilk p	Paired t-test p	Significant at α = 0.05?
Ensemble vs DT	Accuracy	2.2679	+0.0037	0.663	0.0859	No
Ensemble vs DT	F1-Score	0.8171	+0.0022	0.626	0.4597	No
Ensemble vs SVM	Accuracy	4.6182	+0.0030	0.131	0.0099	Yes
Ensemble vs SVM	F1-Score	4.9397	+0.0052	0.846	0.0078	Yes
SVM vs DT	Accuracy	0.4044	+0.0007	0.353	0.7066	No
SVM vs DT	F1-Score	-0.9803	-0.0031	0.898	0.3824	No

The statistical analysis indicates that the ensemble's improvements over the Decision Tree were not statistically significant for either accuracy or F1-score. Therefore, although the ensemble achieved slightly higher mean performance, the observed differences were not sufficiently large relative to the variation across folds to establish a statistically significant improvement.

In contrast, the ensemble significantly outperformed the SVM for both accuracy ($p = 0.0099$) and F1-score ($p = 0.0078$). These results provide statistical support for the improvement observed between the ensemble and SVM during cross-validation.

The comparison between the Decision Tree and SVM also produced no statistically significant differences. Overall, the findings suggest that the ensemble's main statistically supported improvement was relative to the SVM, whereas its advantage over the Decision Tree was comparatively small.

Effect sizes ($d_z = t/\sqrt{5}$) were also calculated for each comparison. Against the SVM, these were large (approximately 2.1 for accuracy, 2.2 for F1-score), while against the Decision Tree they were considerably smaller (approximately 1.0 for accuracy, 0.4 for F1-score). This indicates that while the ensemble's improvement over the SVM was both statistically significant and comparatively large in standardised terms, the absolute differences remained under one percentage point in all cases, meaning even the significant results represent a modest practical improvement.

5.7 SHAP-Based Explainability and Feature Importance

SHAP analysis was used to investigate both global feature importance and individual predictions for the Decision Tree, SVM and ensemble. The same 300 held-out test observations were explained for all three models, allowing their feature importance patterns to be compared directly. Because the SVM produced a decision-function output rather than probability output, SHAP values were normalised to percentages of mean absolute SHAP importance so that relative importance could be compared across the models.

Across the three models, systolic blood pressure, age, hypertension and cholesterol consistently appeared among the most influential variables. Systolic blood pressure had the highest average relative importance across the models at 45.33%. For the Decision Tree, systolic blood pressure accounted for 63.87% of normalised mean absolute SHAP importance, followed by age at 18.94% and cholesterol at 11.58%. The SVM distributed importance more broadly, with hypertension at 29.36%, systolic blood pressure at 27.20%, age at 13.66%, cholesterol at 11.90% and BMI at 5.82%. For the ensemble, systolic blood pressure was again the most influential variable at 44.94%, followed by hypertension at 16.87%, age at 16.30%, cholesterol at 11.54% and BMI at 3.40%.

The dominance of blood-pressure-related variables is broadly consistent with the risk factors discussed in the literature review and the patterns observed during exploratory data analysis. In the original correlation analysis, systolic blood pressure had the strongest correlation with the CVD target at 0.428, followed by diastolic blood pressure at 0.340. Age and cholesterol also demonstrated positive relationships with the target. The SHAP analysis therefore provides model-specific evidence that these variables were important to the predictions produced by the final models.

The SHAP dependence analysis also indicated a nonlinear relationship between systolic blood pressure and the ensemble's output. Lower systolic blood pressure values generally produced negative SHAP contributions, while contributions became increasingly positive at higher values, particularly from approximately 130–140 mmHg onwards. This demonstrates that the ensemble did not treat the relationship between blood pressure and predicted CVD risk as simply linear.

The local SHAP explanations further demonstrated how multiple variables contributed simultaneously to individual predictions. For a correctly classified CVD case, the ensemble predicted a CVD probability of 0.8667 compared with a baseline of approximately 0.497. Systolic blood pressure contributed approximately +0.23, age +0.09 and hypertension +0.05, illustrating how the combination of several features increased the predicted probability. Conversely, for a correctly classified No-CVD case, the predicted probability was 0.1340. The patient's systolic blood pressure contributed approximately −0.25 and younger age approximately −0.07, outweighing the smaller positive contribution from the hypertension indicator.

These local explanations demonstrate the value of SHAP because they show that the model does not base its predictions on one feature in isolation. Instead, predictions result from the combined contributions of multiple variables. However, SHAP explanations describe the behaviour of the trained models and should not be interpreted as evidence that a particular variable causally produces cardiovascular disease.

An important consideration is that the hypertension variable was itself derived from systolic and diastolic blood pressure. Therefore, the high importance of both blood pressure and hypertension should not be interpreted as four independent clinical signals. Instead, their combined importance indicates that blood-pressure-related information played a major role in the predictions. Similarly, the relatively low SHAP importance assigned to variables such as smoking and physical activity does not imply that these factors are clinically unimportant; it only indicates that they contributed less to predictions within this particular dataset and modelling pipeline.

5.8 Overall Discussion

The overall results demonstrate that the machine learning pipeline produced consistent predictive performance across the individual and ensemble models. The Decision Tree and SVM achieved almost identical test accuracy, but their different precision and recall profiles demonstrated that they made somewhat different classification errors.

Feature engineering produced a small improvement over the original feature representation, while feature selection retained comparable accuracy and achieved the highest cross-validation F1-score using only 10 variables. This suggests

that reducing redundancy was beneficial for producing a more compact and interpretable model without sacrificing predictive performance.

The ensemble achieved the highest test accuracy, ROC-AUC, precision, MCC, and Cohen's Kappa among the three final models. However, the statistical analysis showed that its improvement over the Decision Tree was not statistically significant, while its improvements over the SVM were statistically significant for both accuracy and F1-score. Therefore, the ensemble should not be described as universally superior to both individual models. Instead, the results indicate that its performance advantage was most clearly demonstrated relative to the SVM.

The SHAP analysis provided an additional layer of interpretability. Blood pressure, age, hypertension, and cholesterol were consistently among the most influential variables across models. This broad agreement with known cardiovascular risk factors provides useful support for the interpretability of the models, while the model-specific differences in feature importance demonstrate the value of examining explanations across multiple algorithms.

5.9 Limitations

Several limitations should be considered when interpreting these findings. First, the study used a single Kaggle cardiovascular disease dataset. The resulting performance therefore reflects the characteristics and distributions of that dataset and may not generalise directly to other populations or clinical settings.

Second, the dataset is observational, meaning that the models identify statistical patterns rather than causal relationships. SHAP explanations should therefore be interpreted as explanations of model behaviour rather than clinical causal effects.

Third, the hypertension feature is derived directly from systolic and diastolic blood pressure. Consequently, its importance partly overlaps with information already represented by the original blood-pressure variables. This should be considered when interpreting feature importance results.

In addition, the statistical tests used only five cross-validation folds. Because the training sets overlap across folds, the fold scores are not fully independent and the tests have limited power, so the p-values should be read as indicative rather than definitive.

Finally, the ensemble and individual models achieved moderate rather than near-perfect predictive performance. Although the ensemble improved several evaluation metrics, false negatives remained present, demonstrating that the model should not be interpreted as a replacement for professional clinical assessment.

5.10 Future Work

Future work could investigate additional machine learning and ensemble approaches, including alternative boosting methods and more extensive hyperparameter optimisation. External validation using an independent cardiovascular dataset would also be valuable for assessing whether the observed performance generalises to other populations.

Further feature engineering could incorporate additional clinically relevant information where available, such as family history, medication history, dietary factors, or more detailed measures of cardiovascular health. Future studies could also investigate calibration to determine whether predicted probabilities correspond closely to observed disease probabilities.

From an explainability perspective, future work could compare SHAP with additional interpretability methods and investigate whether explanations remain stable across different datasets and model configurations. Finally, testing the models in a realistic clinical decision-support setting would be necessary before considering practical deployment.

6. Conclusion

This study presented an end-to-end, interpretable machine learning pipeline for cardiovascular disease (CVD) risk prediction utilising the Kaggle Cardiovascular Disease dataset (~70,000 patient records). By addressing common clinical machine learning limitations, namely model opacity, feature redundancy, and a lack of statistical rigour in ensemble evaluation, the research demonstrated how data-driven systems can deliver reliable and transparent diagnostic support.

The experimental pipeline demonstrated that deliberate data preprocessing and feature selection can reduce model complexity without sacrificing predictive power. Screened for multicollinearity, the original variable set was refined to 10 optimal features, including clinically meaningful engineered indicators such as Body Mass Index and a hypertension indicator, achieving comparable accuracy and improved cross-validation F1-scores. In baseline evaluations on the held-out test set, the tuned Decision Tree (depth = 5) and Support Vector Machine (SVM with RBF kernel) displayed complementary strengths: the Decision Tree achieved higher sensitivity/recall (67.17%), whereas the SVM demonstrated superior specificity (81.08%) and precision (77.37%).

To leverage these complementary classification error patterns, an equal-weight Soft Voting Ensemble was developed. The ensemble achieved the highest overall performance on the test set across multiple metrics, reaching an accuracy of 73.89%, precision of 77.47%, ROC-AUC of 0.8015, Matthews Correlation Coefficient (MCC) of 0.4816, and Cohen's Kappa of 0.4770. Crucially, stratified 5-fold cross-validation paired t-tests confirmed that the Soft Voting Ensemble achieved statistically significant performance improvements over the standalone SVM classifier in both mean accuracy (+0.30 percentage points, $p = 0.0099$) and F1-score (+0.52 percentage points, $p = 0.0078$), confirming a statistically significant improvement at $\alpha = 0.05$. However, the improvement over the Decision Tree was not statistically significant (accuracy $p = 0.0859$, F1-score $p = 0.4597$), so the ensemble's advantage is best supported relative to the SVM.

To bridge the interpretability gap inherent in complex models, SHapley Additive Explanations (SHAP) were integrated. Global SHAP analysis confirmed that blood pressure indicators (systolic blood pressure ap_hi and the engineered hypertension indicator), age, and cholesterol were the primary drivers of predictions across all models. Furthermore, local SHAP explanations demonstrated how individual patient risk assessments reflect multi-feature interactions rather than isolated single variables, providing human-readable justifications aligned with established cardiovascular risk factors.

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Appendices

Appendix A: AI Declaration

Generative AI and writing assistance tools were used in a supporting role during this project. All outputs were reviewed by the group, and the authors remain responsible for the content, code, and results presented in this report.

Grammarly was used for improving sentence structure and clarity in the report text.

ChatGPT was used to assist the development and optimisation of code and to refine report grammar for clarity.

Gemini was used for assistance with drafting, refining the discussion, and troubleshooting errors in Google Colab.

Claude (Anthropic) was used for proofreading and consistency checks of the report and cross-checking reported figures against notebook outputs.

All models were trained and evaluated by the group in Google Colab. The reported results are taken directly from the notebook outputs and were not generated or altered by AI tools.

Appendix B: Individual Contributions

The table below summarises each member's responsibilities in Phase 2 and Phase 3 and their percentage contributions. All responsibilities listed above were completed by the named member.

Name	Phase 2: Responsibilities	Phase 3: Responsibilities	Contribution (%)
Varushka Dayal	EDA + Feature Engineering	Abstract, Results & Discussion, Conclusion	25
Rahil Sha	Literature Review + Introduction (update) + Remaining Work	Statistical Analysis (code + writing), Experimental Setup section	25
Sasthi Naidoo	Load Dataset + Data Preprocessing + Dataset Description	SHAP code, Explainable AI section	25
Shaista Naicker	Feature Selection + Model Implementation + Preliminary Results	Ensemble Model (code + writing), Model Development section	25